

Problems Caused by Semantic Drift in WordNet Synset Construction

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Abstract—In this study, we summarize the semantic drift problem that occur in specific synsets of KeNet, a Turkish WordNet, which is caused by mis-merging of semantically-related lexical items, morphological markings and false part of speech (POS) matchings. We present our approach to these problems in order to eliminate the semantic drift. We have re-analyzed the dictionary definitions of the items, placed those that possess different verbal markings into separate synsets, and divided synsets based on the POS of the items in them.

Keywords—Turkish wordnet, Kenet, Semantic Drift

I. INTRODUCTION

Since the introduction of the first WordNet in English, which draws comprehensive relations between words and their senses in terms of synonymy, hypernymy, hyponymy, antonymy, and many other relations, two Turkish WordNets have been so far constructed: BalkaNet and KeNet.

The main lexical source for KeNet is the Contemporary Dictionary of Turkish (CDT) published online and in paper by Turkish Language Institute (TLI). The online version of CDT contains 65,944 literal entries. CDT contains some Multi-Word Expression (MWE) entries that have idiomatic senses. The CDT has a pretty straightforward structure without any special mark-ups. For example, synonyms are not marked up but are instead embedded within the sense definitions.

With the help of these embedded sense definitions, synsets in KeNet are formed. One crucial drawback of such embedded sense definitions is the semantic drift occurred in certain synsets. Some synsets contained lexical items that are not in a true synonymy relation. In this work, we have attempted to split these synsets into smaller synsets that include true synonyms. We minimized the effect of semantic drift in order to obtain synsets which carry a single sense.

Creating synsets with the proper items in them is not an easy task. Several reasons can be argued to contribute to this difficulty. One of them is to include all the correct items with the closely-related meanings in all the synsets. To decide on the correct semantic-relatedness of the items is challenging as it requires strict boundaries among the items, which might not be easy to detect in a given language due to the semantic drift between the words. Another difficulty that arises as a result of this semantic drift is the mis-merging of morphologically-related items. This second source of difficulty is the task of separating the synsets based on their morphology correctly. Whereas some morphological derivational processes seem to yield words that could belong to the same synsets as the roots, some do not.

This paper is organized as follows: in Section II, we introduce Turkish wordnets and give brief information about

the main difficulties in Synset construction in the literature. Section III describes our methodology and the problems we have encountered. Lastly, we conclude in Section IV.

II. LITERATURE REVIEW

The first Turkish WordNet is BalkaNet [1], where the researchers automatically extracted synonyms, antonyms and hyponyms from a monolingual Turkish dictionary. The most comprehensive Turkish WordNet is Kenet, which has 80,000 synsets covering 110,000 word-sense pairs [2].

The most important difficulty in synset construction is caused by the lack of clear-cut distinctions between the parts of speech of words that are combined in the same synsets due to their semantic overlap. Regarding the POS categories, we observe this difficulty of differentiating the correct POS of words in some combinations of categories in Turkish, such as adjectives and nouns, adjectives and adverbs and nouns and verbs. The adjective-noun continuum in Turkish has also been discussed in the literature. Regarding that continuum, determining the POS of the items and classifying accordingly is a difficult task, as pointed out by [3]. This is because in Turkish, most adjectives can function as nouns, which makes the split process challenging.

Braun and Haig [3] show that adjectives and nouns in Turkish cannot easily be categorized as separate word classes. They contend that particular nouns may occur in environments that are meant for adjectives. In their analyses, instead of dispensing with the category “adjective” altogether in Turkish, they come up with a continuum based on a statistical analysis of adjective-noun classifications. In this continuum, items are either more “noun”, or “adjective-like” or “neutral”.

Braun and Haig [3] conclude that there is no strict distinction between the categories “adjective” and “noun”. In line with these explanations for adjectives and nouns in the literature, for the source of difficulty caused by the continuum of the POS of the items, we will focus on the difficulties occurred in differentiating synsets including adjectives and nouns. In addition to that, a second continuum we have observed in our analyses and will discuss below is the continuum between adjectives and adverbs.

III. THE TASK OF SPLITTING SYNSETS IN KENET

In this section, we will discuss three main cases we have encountered in splitting the synsets in Turkish KeNet that are mis-merged due to their semantic overlap. Before moving on to the reasons of these mergings, we must briefly mention how we analyzed our synsets. For this task, we displayed all these mis-merged synsets that we have sorted from our

Table I
MEANING-RELATED DRIFT IN VERBS

Merged Id	Literal(s)	SynSet Id	Definition
TUR10-1155530	affetmek unutmak	TUR10-0009720	Herhangi bir kötü davranış için ceza vermekten vazgeçmek; bağışlamak
	hibe etmek teberru etmek	TUR10-1049750	Bir mal veya hakkı karşılık beklemeden birine vermek; bağışlamak

Table II
MEANING-RELATED DRIFT IN ADJECTIVES

Merged Id	Literal(s)	SynSet Id	Definition
TUR10-0054930	ateşin ateşli coşkulu hararetili heyecanlı coşkunca vectli koyu nefes kesici deli huruşan	TUR10-0054930	Coşkusu olan Heyecan veren Azgın
	azgın celalli coşkun	TUR10-0065120	Coşmuş, taşmış
	derin	TUR10-0196230	Ayrıntıya önem verilerek hazırlanan
	lirik	TUR10-0512300	Coşkun, ilhamla dolu

Table III
ENGLISH-TURKISH INCORRECT MAPPING

Merged Id	Literal(s)	SynSet Id	Definition
TUR10-0353710	akak	TUR10-0017110	İrmak, dere, çay, küçük akarsu
	çay	TUR10-1245950	Dereden büyük, ırmaktan küçük akarsu
	öz	TUR10-0607380	Dere, çay
	dere	TUR10-0195670	Genellikle yazın kuruyan küçük akarsu
	ırmak nehir	TUR10-0353710	Çoğunlukla denize dökülen, özellikle genişliği ve taşıdığı su niceliği bakımından en büyük akarsu

previous work with Turkish KeNet on Google Sheets. After creating boundaries to differentiate merged synsets from one another, we analyzed each synset by itself separately. In each synset we split, we created new sets of synsets and wrote new definitions for these new synsets based on the sense differentiations among them. As for the reasons behind these mis-merging of different synsets into a single one, three important ones have been noticed; the continuum created by the meanings of the synsets, the merging of derivationally-related items and the continuum created by the POS of the items. In this section, we will discuss these three reasons in further detail.

A. Meaning-Related Drift

The semantic continuum between lexical items gives rise to synsets that include a huge number of items. When two items are merged together to form a synset, other items included in one of the merged synsets also end up being merged. Such a manner of matching has led to a continuum of senses inside particular synsets, hence merging of senses that are not in fact synonyms of each other and do not form a synset. In other words, some synsets have been merged such that all the senses in them are somehow related but should in fact be arranged as separate synsets. In the example in Table I, in the Turkish dictionary, *hibe etmek* “to donate” is described as “bağışlamak”. This lexical word bağışlamak means both “to donate” and “to forgive”. Annotators mistakenly matched *hibe etmek* and *affetmek* as synonyms since the definition of *affetmek* contains “bağışlamak”, although the meaning of *affetmek* is “to forgive”. Due to this mismatch, we split the synset below into two. One contains the items *hibe etmek* and *teberru etmek* under the sense “to donate”. The other synset is comprised of *affetmek* and *unutmak* meaning “to forgive”.

Another fact observed for the difficulty of splitting the synsets into the correct groups is the semantic continuum between the items. This seems to be the case mostly for the adjective synsets. For instance, in the example given in Table

II, whereas all of the adjectives that are related to one another in terms of their meanings, when we look at the items in the first row (the first newly-divided synset means “excited”) and in the last row (the last newly-divided synset means “lyrical”), we see that they cannot be included in the same synset. In other words, whereas all the items are semantically connected in a way, there need to be some layers between the first and the last items. In our work, by creating boundaries for these inner layers, we split these big synsets into new, smaller synsets, which provided us with the needed semantic drifts.

In the splitting process, we have also observed that some lexical items are merged in the same synset because of the fact that one sense in English can be represented by more than one different sense in Turkish. One instance of such incorrect mapping is given in Table III. In this example, the items such as *dere*, *çay* and *ırmak* are merged under the same synset because these items can be represented under a single item in English such as “stream” or “river”. In our work, we focused on the differentiation of the senses in Turkish and put *dere*, *çay* and *ırmak* into separate synsets.

A more important result of the mis-merging of synsets caused by semantic overlaps is the one which results in the combination of items with different POS categories. Two such examples that we will provide here are the combinations of (i) propositions and nouns and (ii) verbs and nouns.

An example of such a false POS mismatch we found out in a few synsets is the postposition-noun mismatch. Some postpositions appeared inside noun synsets due to semantic similarity. In the synset in Table IV, two different senses of *başka* are included, and these two senses belong to different POS. In the first split synset, *başka* is an adjective and means “another”. The other sense of *başka* comprises its own synset as it is a proposition and no other proposition with a similar meaning is contained in the bigger synset. Both the postposition and the adjective *başka* that were merged in the same synset due to semantic drift have been split into two new synsets.

Table IV
POSTPOSITION-NOUN MIS-MERGING

Merged Id	Literal(s)	SynSet Id	Definition
TUR10-0818600	başka özge	TUR10-0082160	Bilinenden ayrı, değişik, farklı
	başka	TUR10-0082180	Konu edilen, bilinenden ayrı nesne ve kimse için teklik veya çokluk olarak başkası, başkaları biçiminde kullanılan bir söz
	diğer gayrı öbür öteki sair vesair	TUR10-0203010	Bilinenden, sözü edilenden ayrı

Table V
NOUN-VERB MIS-MERGING

Merged Id	Literal(s)	SynSet Id	Definition
TUR10-0456000	gerinmek göğsü kabarmak iftihar etmek kıvançlanmak kıvanmak övünç duymak övünmek	TUR10-0294450	Bir niteliği sebebiyle kendini yücelmiş sayarak bundan abartmalı bir biçimde söz etmek
	hoşnut kalmak hoşnut olmak hoşnutluk duymak memnun olmak	TUR10-1050850	Sevinmek, sevinç duymak, kıvanmak
	kam	TUR10-1246020	Zevk, mutluluk, tat

Table VI
MORPHOLOGY RELATED PROBLEMS: NOUNS

Merged Id	Literal(s)	SynSet Id	Definition
TUR10-0809170	iğrenme iğrenti ikrah ikrahlık irkinti istikrah kerahet kerh nefret tiksınme tiksinti	TUR10-0362890	Bir şeyi tiksindirici bulma
	ürperme	TUR10-0809170	Korku, tiksinti, üşüme vb. yüzünden tüylerin dikilip derinin nokta nokta kabarmasıyla görülen ani titreme

Table VII
MORPHOLOGY RELATED PROBLEMS: ADVERBS

Merged Id	Literal(s)	SynSet Id	Definition
TUR10-0120780	adamakılı bayağı iyiden iyiye evire çevire domuzuna pir güzelce bir temiz	TUR10-0007380	Gereğinden çok, iyice, oldukça, epey
	hıncalınc silme sıvama lebalep ağız ağıza ağzına kadar	TUR10-0341490	Tıka basa dolu olarak, boş yeri kalmayacak biçimde
	az çok berenarı bir dereceye kadar ne kadar nispeten oldukça enikonu hayli epey epeyce epeyi epeyce bir hayli	TUR10-0064910	Az denmeyecek kadar, yetecek kadar

Table VIII
MORPHOLOGY RELATED PROBLEMS: PASSIVE

Merged Id	Literal(s)	SynSet Id	Definition
TUR10-1156950	dayak atmak dövmek kötek atmak kötek çekmek marizlemek sıkıdan geçirmek tekme tokat girişmek tepelemek	TUR10-0889290	Tokat, yumruk, tekme vurarak canını acıtmak
	sopalamak	TUR10-0703060	Sopa ile vurmak, dövmek
	sopalanmak	TUR10-0703080	Sopa ile vurulmak, sövülmek
	sövülmek	TUR10-0706470	Sövme işine konu olmak
	tepelemek	TUR10-0765510	Bozguna uğratmak, hırpalamak

Another example of false POS mismatch we came across with is the noun-verb mismatch. In the example given in Table V, a noun is mistakenly appeared in a verb synset due to its semantic similarity. Due to its different POS, the noun kam “happiness”, which were previously merged into the synset of verbs with the meaning of “to become happy”, was divided and put into a new separate synset.

B. Morphology-Related Problems

Another reason for the mis-merging of synsets is caused by the inclusion of the items that are morphologically related in the same synset. Whereas it does not seem to cause a problem for nouns, adjectives and adverbs, verbs that are morphologically-related are not always supposed to be part of the same synsets. For the former group, nouns, adjectives and adverbs that are derived from the words with the same POS categories belong to the same synset, unless there is semantic differentiation between the root and the derived form. As

it can be seen below, nouns of “iğren-me” and “iğrenti-ti” (Table VI), adjectives of “coşku-lu” and “coşkun-ca” (Table II) and adverbs of “epey-ce” and “epey-i” (Table VII) are put in the same synsets although they are derived forms of the same roots.

On the other hand, verbs that are derived from other verb roots by the addition of morphemes are not usually supposed to be in the same synset. Two such common morphological derivational process for verbs are passivization and causativization. Unlike the case in the previous group, passivized and causativized forms of verbs do not belong to the same synset as they have syntactically-different argument structures in addition to having different meanings. To exemplify, whereas “sopalamak” and “sopala-n-mak” (the latter is the passivized form in Table VIII); and “sınırlamak” and “sınırla-ndır-mak” (the latter is the causativized form Table IX) were mistakenly merged into the same synsets before, we split them into two different synsets because of their semantic

Table IX
MORPHOLOGY RELATED PROBLEMS: CAUSATIVE

Merged Id	Literal(s)	SynSet Id	Definition
TUR10-0206960	diklenmek dikleşmek	TUR10-0204260	Birine karşı ters bir davranışta bulunmak, karşı gelmek, kafa tutmak
	kayıt koymak takyit etmek kayıtlamak	TUR10-1074990	Bağlı kılmak, bir davranışı kısıtlamak, birtakım şartlara bağlamak
	belirlemek sınırlamak çevrelemek	TUR10-0091390	Yeni bir kavramı, özünü oluşturan öğeleri açıklayarak tanımlamak
	hudutlandırmak sınırlamak sınırlandırmak	TUR10-0349580	Sınırını çizmek, sınırını belirtmek veya belirlemek
	gümrük koymak hapsetmek kısıtlamak	TUR10-1040510	Engellemek, sınırlamak
	karışmak	TUR10-0421380	Engellemek, araya girmek
	kısıtlamak	TUR10-0454120	Önceden verilmiş olan hak ve hürriyetlerin sınırlarını daraltmak, tahdit etmek

Table X
NOUN-ADJECTIVE MIS-MERGING

Merged Id	Literal(s)	SynSet Id	Definition
TUR10-0312230	bilek çelim erk	TUR10-0102350	Bir işi yapabilme gücü, kudret, iktidar
	can dirilik	TUR10-0131370	Güç
	fer	TUR10-0268090	Güç, kuvvet, nüfuz
	güç	TUR10-0312170	Ağır ve yorucu emekle yapılan, müşkül
	güç	TUR10-0312230	Büyük etkinliği ve önemi olan nitelik
	erk kuvvet nüfuz	TUR10-0250960	Sözü geçerlik, istediğini yaptırabilme gücü

Table XI
NOUN-ADJECTIVE DIFFERENTIATION

Merged Id	Literal(s)	SynSet Id	Definition
TUR10-0603090	alemdar imam lider önder reis şef tek adam	TUR10-0026380	Gücü, ünü ve toplumsal yeri dolayısıyla, belli zaman ve durumlar içinde, ilişkili bulunduğu küme veya toplumun tutum, davranış ve etkinliklerini değiştirip yönetme yeteneğini gösteren kimse
	lider	TUR10-0511100	Bir yarışmada başta bulunan takım veya yarışmacı
	öncü öncül	TUR10-0603090	Önder, kılavuz

and syntactic differences.

C. POS-Related Drift

The third, and the most crucial, reason observed to cause this wrong merging of synsets was the POS of items. As explained for the case of adjectives and nouns in the previous section, there is not a clear-cut distinction between many parts of speech of words in Turkish. The presence of such a continuum between the POS of words led the annotators to combine synsets which behave, let us say, more like adjectives or nouns, into a single synset since they, together with the layers between them, form a continuum.

In contrast to the differentiations caused by the merging of words with different POS categories due to their semantic overlap as described in Section III.A, the cases that will be described here are a result of the continuum between the POS categories of words. As the differentiation between the POS categories are found to be difficult in these cases, displaying these synsets which include all the items across the continuum have made it possible to decide on the boundaries between the POS categories, and thus, between the new synsets. In this section, we will look at the continuums created by the POS of the items one by one.

1) *Noun-Adjective Differentiation*: The synsets were expected to be formed based on the POS to which lexical items belong. However, as mentioned before, POS differentiations are not always easy in Turkish. The following data in Table X exemplify the adjective-noun continuum. The split synset below contains the item “güç” which can be used both as a noun and as an adjective, as is also defined by the dictionary of Turkish Language Association. Both senses of the item

“güç” are found in the same synset due to their semantic similarity. It means “force” when it is used a noun, and “hard, troublesome” when it is used an adjective. We placed these items into separate synsets based on their POS.

Furthermore, many nouns in Turkish can be used as adjectives and vice versa. The following synset in Table XI contains items that may function as adjectives. The monolingual Turkish dictionary classifies some words that are related to human beings as adjectives and their definition carry a word meaning “person” (“kimse”) in parentheses. This means that the word can function both as an adjective and as a noun. In the synset above, for instance, items such as lider, önder, öncü “leader” could either be an adjective or a noun. We paid attention to the descriptions and categorizations of the dictionary.

2) *Adjective-Adverb Differentiation*: In addition to adjective-noun categorization, adverbs can be replaced by adjectival elements, as well. There is not a strict distinction between the adjective and adverb categories with respect to certain lexical items in Turkish, either. We observed such adjective-adverb confusion in some synsets, as exemplified in Table XII. The synset given below confuses the adjective and adverb classes. If we look at the first newly-formed synset, we will see that it contains adverbs as they are described as “in an x way” in the Turkish dictionary. Bear in mind that those items can function as adjectives, too. The fact that there is no clear distinction between the POS categories of “adjective” and “adverb” in Turkish complicates the splitting process. Considering their dictionary description and their principal usage, we classified them adverbs. We categorized

Table XII
ADJECTIVE-ADVERB MIS-MERGING

Merged Id	Literal(s)	SynSet Id	Definition
TUR10-0503870	ahbapça laubali teklifsizce	TUR10-0014930	Dostça, içten, aşırı samimi bir biçimde
	cüretkar cüretkâr hüremtsiz saygısız	TUR10-0143810	Gereken saygıyı göstermeyen, saygısı olmayan
	gayriciddi	TUR10-0284710	Ciddi olmayan, laubali, ciddiyetsiz
	dili uzun küstah laubali küstahça mektep görmemiş riayetsiz	TUR10-0205700	İncitici sözler söyleyen, kaba, terbiyesiz
	şımarık şopar	TUR10-0730140	Şımarmış, şımartılmış
	vahşi	TUR10-0812710	Kaba, saygısız, uyum sağlayamayan

the rest of the items as adjectives since they are chiefly used to modify nouns. Note, nevertheless, that words such as küstahça “impudent” and şımarık “spoiled” can function as adverbs.

IV. CONCLUSION

In this paper, we have looked at the process of splitting the synsets that are wrongly-merged due to the semantic drift. In addition to the fact that semantic drift by itself poses difficulties in the decision making process of creating the necessary boundaries of the synsets, we have also mentioned that semantic drift may result in two other problems. The first one is the inclusion of morphologically-related items in a single synset. We have observed that semantically related verbs that possess different morphological markings are merged inside the same synset. We placed the verbs with passive and causative marking in separate synsets. The other problem is the inclusion of items that have distinct POS. We distinguished between different POS the lexical items in a

synset belong to and classified them accordingly. We have attempted to put boundaries between adjectives and nouns, adjectives and adverbs, and postpositions and nouns merged under the same synset. Minimizing the effects of semantic drift and incorrect POS matchings by splitting synsets have provided us with synsets that contain true synonyms.

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